

## 24.5 Regression calibration and SIMEX

The moment correction and IV are tied to linear moment equations. Two more portable ideas were designed for logistic, Cox and other nonlinear regressions, where they are approximations. **Regression calibration** replaces the unobserved regressor by its best prediction from what was observed. **SIMEX** watches the naive estimate deteriorate as error is added, and extrapolates back to no error. In the linear model both can be exact.

### 24.5.1 Regression calibration

Error is **nondifferential** if the distribution of  $Y$  given  $(\mathbf{x}, \mathbf{z}, \mathbf{w})$  does not depend on  $\mathbf{w}$ , as for classical error.

#### Proposition 24.5.1. Regression calibration

Suppose  $E(Y | \mathbf{x}, \mathbf{z}) = \alpha + \boldsymbol{\beta}_x^\top \mathbf{x} + \boldsymbol{\beta}_z^\top \mathbf{z}$  and the error is nondifferential, and let  $\mathbf{m}(\mathbf{w}, \mathbf{z}) = E(\mathbf{x} | \mathbf{w}, \mathbf{z})$ .

- (a)  $E(Y | \mathbf{w}, \mathbf{z}) = \alpha + \boldsymbol{\beta}_x^\top \mathbf{m}(\mathbf{w}, \mathbf{z}) + \boldsymbol{\beta}_z^\top \mathbf{z}$ .
- (b) If moreover  $\text{Var}(Y | \mathbf{x}, \mathbf{z}) = \sigma^2$  and  $\text{Cov}(\mathbf{x} | \mathbf{w}, \mathbf{z}) = \mathbf{C}$  does not depend on  $(\mathbf{w}, \mathbf{z})$ , as when  $(\mathbf{x}, \mathbf{z}, \mathbf{u})$  is jointly normal, then  $\text{Var}(Y | \mathbf{w}, \mathbf{z}) = \sigma^2 + \boldsymbol{\beta}_x^\top \mathbf{C} \boldsymbol{\beta}_x$ . The regression of  $Y$  on  $(1, \mathbf{m}(\mathbf{w}, \mathbf{z}), \mathbf{z})$  is then a homoscedastic linear model with the original coefficients.
- (c) Without distributional assumptions, let the error be classical with finite second moments. With  $\mathbf{t} = (1, \mathbf{x}^\top, \mathbf{z}^\top)^\top$ ,  $\mathbf{s} = (1, \mathbf{w}^\top, \mathbf{z}^\top)^\top$ ,  $\mathbf{A} = E(\mathbf{t}\mathbf{t}^\top)$  positive definite and  $\mathbf{S} = E(\mathbf{s}\mathbf{s}^\top)$ , the best linear predictor of  $\mathbf{t}$  from  $\mathbf{s}$  is  $\hat{\mathbf{t}} = \mathbf{A}\mathbf{S}^{-1}\mathbf{s}$ , and the best linear predictor of  $Y$  from  $\hat{\mathbf{t}}$  is  $\boldsymbol{\theta}^\top \hat{\mathbf{t}}$ , where  $\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}_x^\top, \boldsymbol{\beta}_z^\top)^\top$ .
- (d) The sample version of (c) reproduces the method-of-moments estimator exactly. If  $\hat{\mathbf{S}} = n^{-1} \sum_i \mathbf{s}_i \mathbf{s}_i^\top$ ,  $\hat{\mathbf{A}} = \hat{\mathbf{S}} - \mathbf{D}$  for a symmetric matrix  $\mathbf{D}$  with  $\hat{\mathbf{A}}$  nonsingular, and  $\hat{\mathbf{t}}_i = \hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}\mathbf{s}_i$ , then the least squares coefficients of  $y$  on the  $\hat{\mathbf{t}}_i$  are  $\hat{\mathbf{A}}^{-1}n^{-1} \sum_i \mathbf{s}_i y_i$ , which is Eq. 24.3.1 with  $\mathbf{D}_i = \mathbf{D}$ .

*Proof.* (a) By the tower property and nondifferential error,  $E(Y | \mathbf{w}, \mathbf{z}) = E\{E(Y | \mathbf{x}, \mathbf{z}, \mathbf{w}) | \mathbf{w}, \mathbf{z}\} = E\{\alpha + \boldsymbol{\beta}_x^\top \mathbf{x} + \boldsymbol{\beta}_z^\top \mathbf{z} | \mathbf{w}, \mathbf{z}\}$ , which is the claim.

(b) The law of total variance, conditioning on  $\mathbf{x}$  inside  $(\mathbf{w}, \mathbf{z})$ , gives  $\text{Var}(Y | \mathbf{w}, \mathbf{z}) = E\{\text{Var}(Y | \mathbf{x}, \mathbf{z}, \mathbf{w}) | \mathbf{w}, \mathbf{z}\} + \text{Var}\{E(Y | \mathbf{x}, \mathbf{z}, \mathbf{w}) | \mathbf{w}, \mathbf{z}\} = \sigma^2 + \boldsymbol{\beta}_x^\top \mathbf{C} \boldsymbol{\beta}_x$ . Under joint normality the conditional covariance of  $\mathbf{x}$  given  $(\mathbf{w}, \mathbf{z})$  does not depend on the conditioning values (Theorem 3.5.2).

(c) The best linear predictor (Definition 6.11.1, Theorem 2.5.2), with the constant among the predictors, is the projection in  $L^2$  onto the span of the entries of  $\mathbf{s}$ , so it is  $E(\mathbf{t}\mathbf{s}^\top)\mathbf{S}^{-1}\mathbf{s}$ . Under classical error  $E(\mathbf{t}\mathbf{s}^\top) = E\{\mathbf{t}(\mathbf{t} + \tilde{\mathbf{u}})^\top\} = \mathbf{A}$ , and  $\mathbf{A}\mathbf{S}^{-1}\mathbf{s}$  follows. The entries of  $\hat{\mathbf{t}}$  span the same space as those of  $\mathbf{s}$ , since  $\mathbf{A}\mathbf{S}^{-1}$  is nonsingular. Their second-moment matrix is  $\mathbf{A}\mathbf{S}^{-1}\mathbf{S}\mathbf{S}^{-1}\mathbf{A} = \mathbf{A}\mathbf{S}^{-1}\mathbf{A}$ , and  $E(\hat{\mathbf{t}}Y) = \mathbf{A}\mathbf{S}^{-1}E(\mathbf{s}Y) = \mathbf{A}\mathbf{S}^{-1}\mathbf{A}\boldsymbol{\theta}$ , because  $E(\mathbf{s}Y) = \mathbf{A}\boldsymbol{\theta}$  as in Section 24.3. The projection coefficients are  $(\mathbf{A}\mathbf{S}^{-1}\mathbf{A})^{-1}\mathbf{A}\mathbf{S}^{-1}\mathbf{A}\boldsymbol{\theta} = \boldsymbol{\theta}$ .

(d) With the calibrated regressors  $\hat{\mathbf{t}}_i = \hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}\mathbf{s}_i$ , the normal equations (Theorem 6.4.2) have matrix  $n^{-1} \sum_i \hat{\mathbf{t}}_i \hat{\mathbf{t}}_i^\top = \hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}\hat{\mathbf{A}}$  and right side  $\hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}n^{-1} \sum_i \mathbf{s}_i y_i$ . Solving,  $(\hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}\hat{\mathbf{A}})^{-1}\hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}n^{-1} \sum_i \mathbf{s}_i y_i =$

$$\hat{\mathbf{A}}^{-1} n^{-1} \sum_i \mathbf{s}_i y_i.$$

■

The coordinates of  $\hat{\mathbf{t}}$  for the constant and  $\mathbf{z}$  reproduce them, so only the mismeasured regressors are replaced. By (c) and (d), linear regression calibration needs no distributional assumption and *is* the moment correction, with the standard errors of Theorem 24.3.1. Second-stage standard errors, which treat the calibrated values as data, are too small. A **validation subsample** in which  $\mathbf{x}$  is measured identifies the calibration directly, by regressing  $\mathbf{x}$  on  $(\mathbf{w}, \mathbf{z})$  there, and needs only (a): nondifferential error and a correct model for  $E(\mathbf{x} \mid \mathbf{w}, \mathbf{z})$ . In a logistic regression, (a) gives  $E\{G(\alpha + \beta X) \mid w\}$ , not  $G(\alpha + \beta m(w))$ , so calibration is an approximation, good when  $\text{Var}(X \mid W)$  is small (Carroll, Ruppert, Stefanski and Crainiceanu, 2006).

*Remark* (Two meanings of “calibration”). In [Section 12.5](#) *calibration* meant inverse prediction of  $x_0$  from a new response. Regression calibration instead predicts the regressor from its measurement.

### Example. Calibrating the mean reading

The replicates estimate the error variance of the mean reading as 37.04. The calibrated pressure  $\hat{x}_i$  has standard deviation 10.36, against 13.11 for the mean reading: each reading is shrunk towards what age predicts. Regressing  $y$  on  $(1, \hat{x}, \text{age})$  gives 0.0757 and 0.0018, exactly the moment correction of Example (Correcting with two readings), by [Proposition 24.5.1\(d\)](#).

```
import numpy as np

rng = np.random.default_rng(2401)
n = 300
age = rng.normal(55, 10, n)
x = 130 + 0.6 * (age - 55) + rng.normal(0, np.sqrt(108), n)  # long-run blood
                        pressure, sd 12
w = x[:, None] + rng.normal(0, 9, (n, 2))                  # two clinic readings
y = -4 + 0.08 * x + 0.0 * age + rng.normal(0, 1.0, n)      # age has no effect
                        given x

wbar = w.mean(axis=1)
s2_ubar = np.mean((w[:, 0] - w[:, 1]) ** 2) / 4             # error variance of
                        the mean reading
W = np.column_stack([np.ones(n), wbar, age])
S = W.T @ W / n                                            # raw second moments
                        of (1, wbar, age)
Suu = np.diag([0.0, s2_ubar, 0.0])

# calibration: best linear predictor of (1, x, age) from (1, wbar, age), using E(w x
# ^T) = S - Suu
Gamma = np.linalg.solve(S, S - Suu)
x_hat = (W @ Gamma)[:, 1]                                 # predicted long-run
                        pressure
b_rc = np.linalg.lstsq(np.column_stack([np.ones(n), x_hat, age]), y, rcond=None)[0]
b_mom = np.linalg.solve(S - Suu, W.T @ y / n)             # method of moments,
                        Section 24.3
print("regression calibration:", b_rc.round(4))
```

```
print("method of moments:      ", b_mom.round(4))
```

[Run this code online](#)

## 24.5.2 SIMEX

Cook and Stefanski (1994) treat the naive estimator as a black box. Data cannot be made more accurate, but they can be made *less* accurate by adding simulated error; watching the naive estimate move as the error variance grows, we extrapolate back to zero. Let  $\sigma_u^2$  be the known or estimated error variance of  $W$ .

1. **Simulation.** For each  $\zeta$  in a grid such as  $\{0.5, 1, 1.5, 2\}$  and  $b = 1, \dots, B$ , form remeasured data  $W_{b,i}(\zeta) = W_i + \sqrt{\zeta} \sigma_u V_{b,i}$ , with  $V_{b,i} \stackrel{\text{iid}}{\sim} N(0, 1)$  independent of the data, and compute the naive estimate  $\hat{\theta}_b(\zeta)$  from  $(W_b(\zeta), \mathbf{z}, Y)$ . Average over  $b$  to get  $\hat{\theta}(\zeta)$ , and put  $\hat{\theta}(0)$  equal to the naive estimate.
2. **Extrapolation.** Fit a function  $g(\zeta; \gamma)$  of  $\zeta$  to the points  $(\zeta, \hat{\theta}(\zeta))$ , one coefficient at a time, and report  $g(-1; \hat{\gamma})$ .

The remeasured error  $U + \sqrt{\zeta} \sigma_u V$  is still classical, with variance  $(1 + \zeta) \sigma_u^2$ , so  $\zeta = -1$  means no error. Averaging over  $b$  removes simulation noise only.

### Proposition 24.5.2. SIMEX in the linear model

Assume the conditions of Theorem 24.1.1(c), with one mismeasured scalar regressor of error variance  $\sigma_u^2$ , and fix  $\zeta \geq 0$ . For each  $b$ , the naive least squares coefficients from the remeasured data converge with probability one to  $\mathbf{g}(\zeta) = (g_x(\zeta), \mathbf{g}_z(\zeta))$ , where

$$g_x(\zeta) = \frac{\beta_x \sigma_{x \cdot z}^2}{\sigma_{x \cdot z}^2 + (1 + \zeta) \sigma_u^2}, \quad \mathbf{g}_z(\zeta) = \boldsymbol{\beta}_z + \frac{(1 + \zeta) \sigma_u^2}{\sigma_{x \cdot z}^2 + (1 + \zeta) \sigma_u^2} \beta_x \boldsymbol{\pi}, \quad (24.5.1)$$

and so does their average over  $b$ . Moreover:

- (a)  $g_x(-1) = \beta_x$  and  $\mathbf{g}_z(-1) = \boldsymbol{\beta}_z$ ;
- (b) every coefficient has the form  $a + b/(c + \zeta)$  with  $c = 1 + \sigma_{x \cdot z}^2/\sigma_u^2$ ; in particular  $1/g_x(\zeta)$  is linear in  $\zeta$  when  $\beta_x \neq 0$ ;
- (c) if a polynomial of fixed degree is fitted by least squares to the points  $(\zeta_k, \hat{\theta}(\zeta_k))$  on a fixed grid, the SIMEX estimate converges to the value at  $-1$  of the polynomial fitted in the same way to  $(\zeta_k, \mathbf{g}(\zeta_k))$ , which in general differs from  $\mathbf{g}(-1)$ .

*Proof.* The remeasured data satisfy Definition 24.1.1 with error  $U + \sqrt{\zeta} \sigma_u V$ , whose variance is  $(1 + \zeta) \sigma_u^2$ , and the observations  $(X_i, \mathbf{z}_i, U_i, V_{b,i}, \varepsilon_i)$  are independent over  $i$ . By Theorem 24.1.1(c), with  $\sigma_u^2$  replaced by  $(1 + \zeta) \sigma_u^2$ , the coefficients converge to  $\lambda_\zeta \beta_x$  and  $\boldsymbol{\beta}_z + (1 - \lambda_\zeta) \beta_x \boldsymbol{\pi}$ , where  $\lambda_\zeta = \sigma_{x \cdot z}^2 / \{\sigma_{x \cdot z}^2 + (1 + \zeta) \sigma_u^2\}$ . This is Eq. 24.5.1. A finite average of sequences converging with probability one converges to the average of the limits. (a) At  $\zeta = -1$ ,  $\lambda_\zeta = 1$ . (b) The denominator is  $\sigma_u^2(c + \zeta)$ . So  $g_x(\zeta) = (\beta_x \sigma_{x \cdot z}^2 / \sigma_u^2) / (c + \zeta)$ , and  $(1 + \zeta) / (c + \zeta) = 1 - (c - 1) / (c + \zeta)$  gives  $\mathbf{g}_z(\zeta) = \boldsymbol{\beta}_z + \beta_x \boldsymbol{\pi} - (c - 1) \beta_x \boldsymbol{\pi} / (c + \zeta)$ . (c) On a fixed grid the fitted polynomial's coefficients are a fixed linear function of the fitted points (the least squares solution with a fixed model matrix), so its value at  $-1$  is a continuous function of the points, and the points converge.

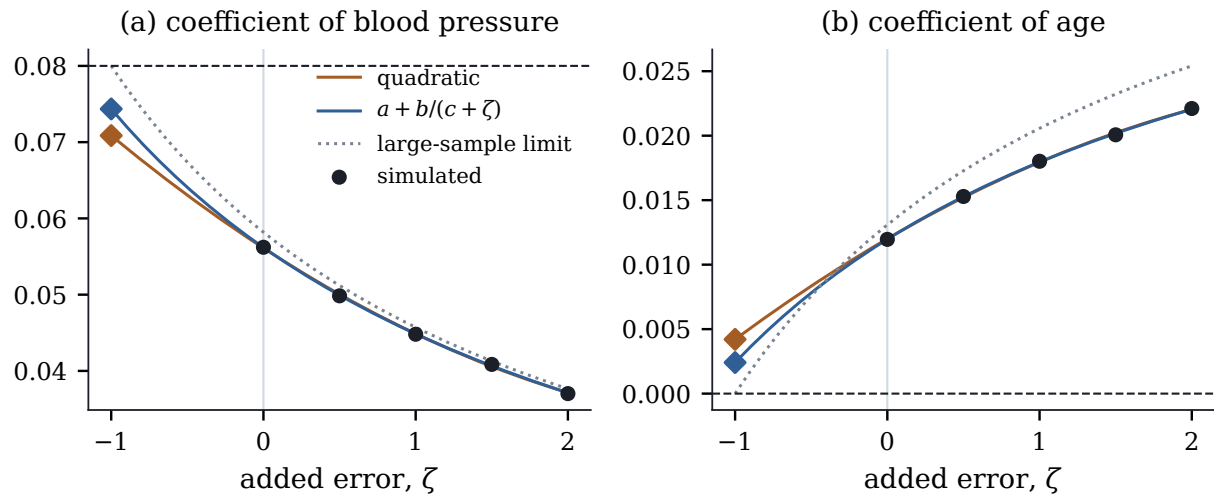


Figure 24.5.1: SIMEX: averaged naive coefficients (dots), quadratic and rational extrapolants with their values at  $\zeta = -1$  (diamonds), the limits Eq. 24.5.1 (dotted) and the truth (dashed).

So the rational form matches the limit exactly, and the rational SIMEX estimate is consistent provided the nonlinear fit is unique and continuous in the points, which we do not verify. The quadratic, a common default because it is linear in its parameters, recovers only part of the attenuation (Exercise (B1)). With several mismeasured regressors, or in nonlinear models, no simple extrapolant is exact and SIMEX is only approximately consistent. Its appeal is that it needs nothing but the naive estimator.

#### Example. SIMEX for the blood-pressure study

Apply SIMEX to the regression on the mean reading and age, with error variance 37.04, grid  $\{0, 0.5, 1, 1.5, 2\}$  and  $B = 200$  (Figure 24.5.1). Extrapolating to  $\zeta = -1$ :

| Coefficient       | naive  | SIMEX,<br>quadratic | SIMEX,<br>$a + b/(c + \zeta)$ | moment<br>correction | truth |
|-------------------|--------|---------------------|-------------------------------|----------------------|-------|
| blood<br>pressure | 0.0562 | 0.0709              | 0.0744                        | 0.0757               | 0.08  |
| age               | 0.0120 | 0.0042              | 0.0024                        | 0.0018               | 0     |

The rational extrapolant nearly reproduces the moment correction; the quadratic stops short, and by Proposition 24.5.2(c) its large-sample limit is 0.0749, only 0.77 of the way from the naive limit 0.0582 to 0.08, and 0.0030 for age. A pairs bootstrap (Definition 23.1.1) over patients, re-estimating the error variance and rerunning SIMEX ( $B = 50, 200$  resamples), gives standard error 0.0076 for the rational SIMEX slope, as for the moment correction.

```
from scipy.optimize import curve_fit

def naive_fits(wcol, age, y):
    """Least squares coefficients of y on (1, wcol, age) for a stack of remeasured
    columns."""
    m, k = wcol.shape
```

# m remeasured data

```

sets of size k
Xs = np.stack([np.ones((m, k)), wcol, np.broadcast_to(age, (m, k))], axis=2)
XtX = np.einsum("mki,mkj->mij", Xs, Xs)
Xty = np.einsum("mki,k->mi", Xs, y)
return np.linalg.solve(XtX, Xty[:, :, None])[:, :, 0]

def simex(wbar, age, y, s2_ubar, zetas, B, rng):
    """Average naive coefficients after adding error of variance zeta * s2_ubar."""
    out = [naive_fits(wbar[None, :], age, y)[0]]
    for z in zetas[1:]:
        extra = rng.normal(0, np.sqrt(z * s2_ubar), (B, len(y)))
        out.append(naive_fits(wbar + extra, age, y).mean(axis=0))
    return np.array(out) # one row per zeta

def rational(z, a, b, c):
    return a + b / (c + z)

zetas = np.array([0.0, 0.5, 1.0, 1.5, 2.0])
G = simex(wbar, age, y, s2_ubar, zetas, 200, np.random.default_rng(2450))
est_quad, est_rat = [], []
for j in [1, 2]: # coefficient of x, of
    age
    est_quad.append(np.polyval(np.polyfit(zetas, G[:, j], 2), -1.0))
    p, _ = curve_fit(rational, zetas, G[:, j], p0=[0.0, G[0, j], 1.5], maxfev=20000)
    est_rat.append(rational(-1.0, *p))
print("naive (zeta = 0): ", G[0, 1:].round(4))
print("SIMEX, quadratic: ", np.round(est_quad, 4))
print("SIMEX, rational: ", np.round(est_rat, 4))

```

[Run this code online](#)

### 24.5.3 Which correction?

In the linear model the moment correction and regression calibration coincide, rational SIMEX agrees up to simulation noise, and IV trades precision for not needing an error variance. Beyond it, only regression calibration and SIMEX carry over. All rest on information from outside the regression of  $Y$  on  $W$ . Without it, Proposition 24.2.2 is the honest summary: a range of slopes, with the naive estimate at one end.

### 24.5.4 Exercises

#### 24.5.4.1 A. Check your understanding

##### Exercise. A1

In simple regression without  $\mathbf{z}$ , show directly that regression calibration ([Proposition 24.5.1](#)) with the linear calibration  $\hat{x} = \bar{w} + \hat{\lambda}(w - \bar{w})$ ,  $\hat{\lambda} = (s_{ww} - \sigma_u^2)/s_{ww}$ , gives the slope  $\hat{\beta}/\hat{\lambda}$ , where  $\hat{\beta}$  is the naive slope.

#### 24.5.4.2 B. Practice

**Exercise. B1**

In simple regression with reliability  $\lambda$ , write  $g(\zeta) = \lambda\beta/\{1 + \zeta(1 - \lambda)\}$ . Show that the quadratic through  $(\zeta, g(\zeta))$  for  $\zeta = 0, 1, 2$  takes the value  $3g(0) - 3g(1) + g(2)$  at  $\zeta = -1$ . Evaluate it for  $\lambda = 0.5$  and  $\lambda = 0.64$ , as fractions of  $\beta$ .

*Solution.* The quadratic through three equally spaced points is determined by the forward differences. Written as  $p(\zeta) = g_0 + \zeta\Delta g_0 + \frac{1}{2}\zeta(\zeta - 1)\Delta^2 g_0$ , it gives  $p(-1) = g_0 - \Delta g_0 + \Delta^2 g_0 = g_0 - (g_1 - g_0) + (g_2 - 2g_1 + g_0) = 3g_0 - 3g_1 + g_2$ . With  $\lambda = 0.5$ :  $g_0 = \beta/2$ ,  $g_1 = \beta/3$ ,  $g_2 = \beta/4$ , so  $p(-1) = \beta(3/2 - 1 + 1/4) = 0.75\beta$ . With  $\lambda = 0.64$ :  $g_0 = 0.64\beta$ ,  $g_1 = 0.64\beta/1.36$ ,  $g_2 = 0.64\beta/1.72$ , so  $p(-1) = \beta(1.92 - 1.4118 + 0.3721) \approx 0.880\beta$ . The quadratic removes most of the attenuation but not all of it, and it does worse as the reliability falls.

**Exercise. B2**

A validation subsample of  $n_v$  patients has the true  $x$  measured along with  $(w, z)$ , and the main sample has only  $(w, z, y)$ . Describe regression calibration using the validation data. Which of the assumptions of [Proposition 24.5.1](#) does it need, and which does it avoid? Why does using the validation patients'  $x$  directly in the outcome regression not help if their  $y$  is not recorded?

**24.5.4.3 C. Going deeper****Exercise. C1**

Simulate a logistic regression  $\Pr(Y = 1 | X) = 1/(1 + e^{-(\alpha + \beta X)})$  with  $X \sim N(0, 1)$ ,  $\beta = 1$ ,  $n = 2000$ , and a reading  $W = X + U$  with  $\sigma_u^2 = 0.5$ . Compare the naive estimate, regression calibration ([Proposition 24.5.1](#): replace  $W$  by  $E(X | W)$ ), and SIMEX with quadratic and rational extrapolants, over repeated samples. Which is closest to  $\beta$ ?